

Carrier Recovery

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1 Introduction

1.1 Frequency Recovery

1.1.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4} + m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i + \theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i + 4\theta[i])} + n'[i] \quad (1)$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i + \theta[i])} + n[i], \quad (2)$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$ which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4\arg(x[i])\}|_{-\pi}^{\pi} \quad (3)$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \quad (4)$$

for data-aided operation where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

1.1.2 Rife and Boorstyn (FFT Search)

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \dots, z[i+L]$ where L is the observation length. The Rife and Boorstyn (R&B) algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give

$$\delta = -\Re \left[\frac{X_{k+1} - X_{k-1}}{2x_k - X_{k-1} - X_{k+1}} \right]. \quad (5)$$

The updated index $k' = k + \delta$, which is then scaled by the sampling rate to find the true CFO. The CPON specification allows for 11 training symbols at the start of each subframe. The resolution in k is determined by the FFT size N_{FFT} as $\epsilon(k) = 1/2N_{FFT}$. For data-aided operation, only the 11 training symbols are available. This would give a resolution of $\epsilon(k) \approx 0.05$. In terms of the estimated frequency $\hat{f}_d$ and symbol rate R_s , $\epsilon(\hat{f}_d) \approx 0.05R_s$. This would likely be too large for the downstream phase recovery to correct, so a larger FFT can be used with zeros padded at the end of the observed sequence. The normalised mean squared-error (MSE) in the estimated CFO is shown in ?? for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarizations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) gives a lower bound on the achievable MSE and is given by

$$MCRB = \frac{3}{2\pi^2 L^3 \times SNR}. \quad (6)$$

R&B achieves this bound for SNRs over ≈ 4 dB (the threshold SNR). At lower SNRs, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

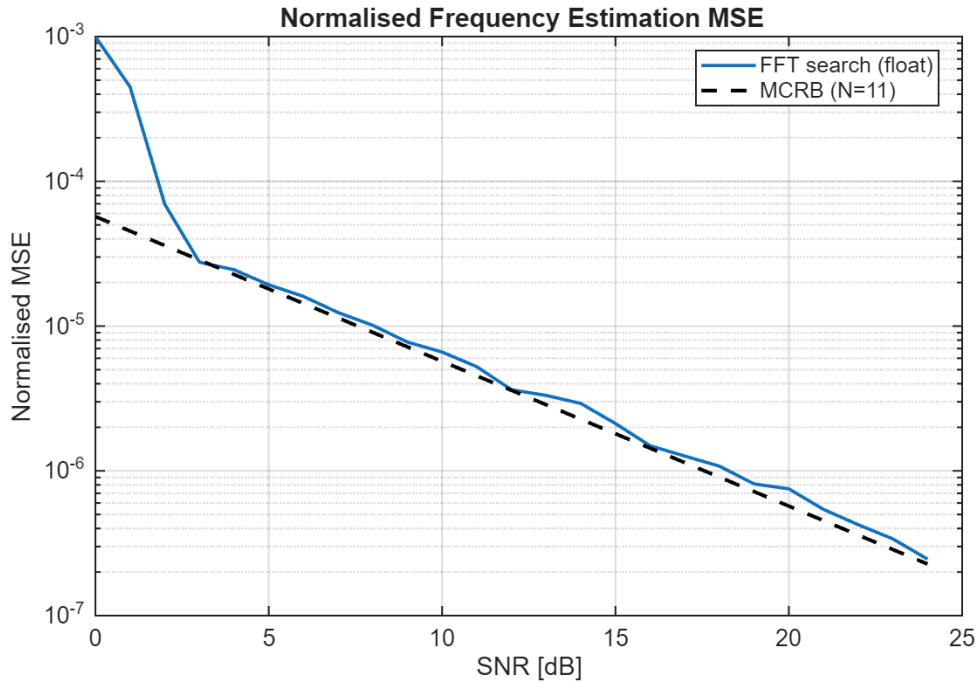


Figure 1: Normalised MSE of the R&B algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the R&B algorithm improves with longer observation lengths, as shown in ?? for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

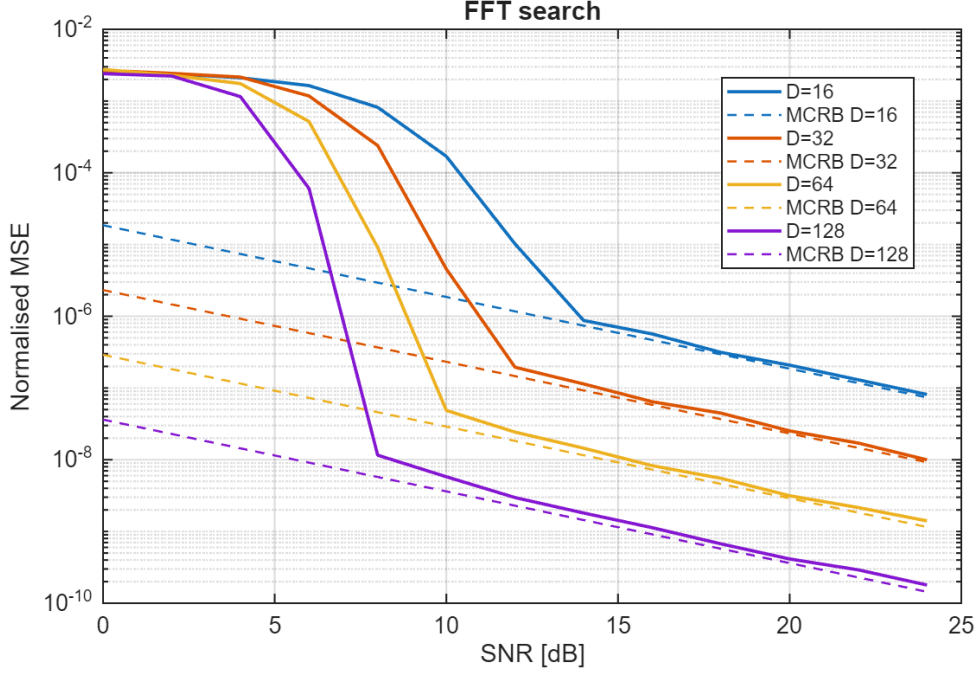


Figure 2: Normalised MSE of the R&B algorithm for non-data-aided operation and different observation lengths.

1.1.3 Differential Phase + Kay Estimator

The R&B algorithm exhibits good performance even at low SNRs but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2} \log_2(N_{FFT})$ complex multiplications and $N_{FFT} \log_2(N_{FFT})$ complex additions. The fine search from ?? also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \quad (7)$$

over several symbols and averaging to remove the effect of θ . Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$ so exhibits poor accuracy as shown in ??. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi} \sum_{k=1}^L \frac{6k(L-k)}{L(L^2-1)} \arg(z[k]z^*[k-1]). \quad (8)$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from ?? and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. This is demonstrated by considering the phase of symbols at either end of the observation sequence. The phase drift between these symbols $\Delta\phi = 2\pi f_d L T_s$. If $|\Delta\phi| > 2\pi$ then a phase wrap is guaranteed to occur, giving an estimated phase drift $\Delta\hat{\phi} = \Delta\phi \pm 2\pi$ and an estimated CFO $\hat{f}_d = f_d \pm \frac{1}{L T_s}$. Thus, a first-pass CFO correction is needed to bring the residual CFO within range for the Kay estimator.

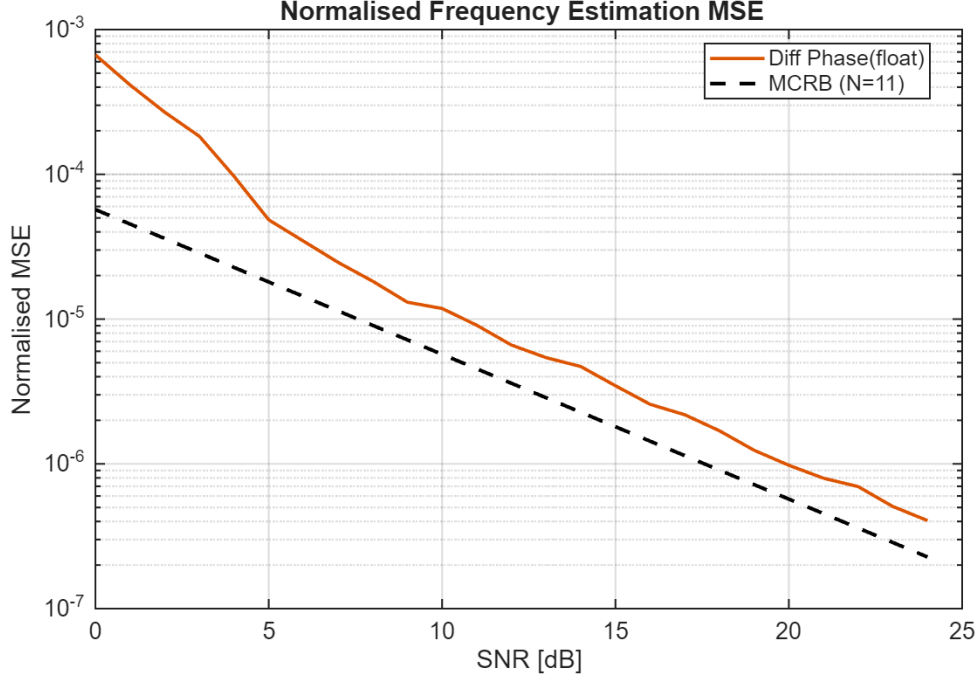


Figure 3: Normalised MSE of frequency estimation with data-aided differential phase for $L = 11$ training symbols.

The performance of the Differential Kay (DK) estimator for data-aided operation is shown in ???. The MCRB is achieved above an SNR of ≈ 7 dB. This is higher than for R&B but at a reduced computational cost. Unlike the R&B algorithm, the threshold at which the MSE of the DK estimator deviates from the MCRB does not depend on observation length, as shown in ???.

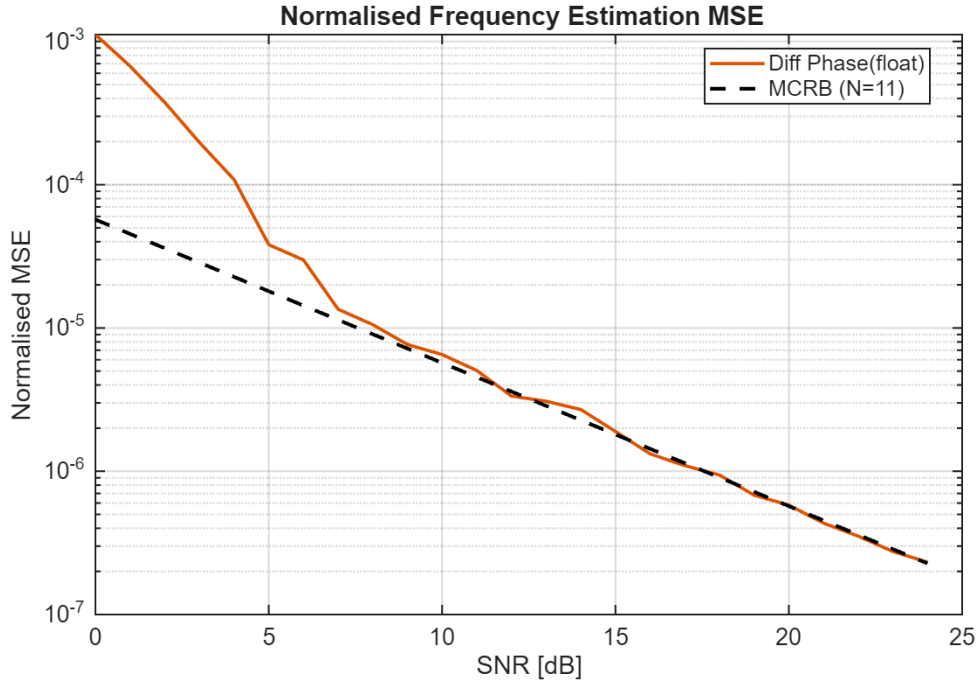


Figure 4: Normalised MSE of frequency estimation with the DK estimator for $L = 11$ training symbols.

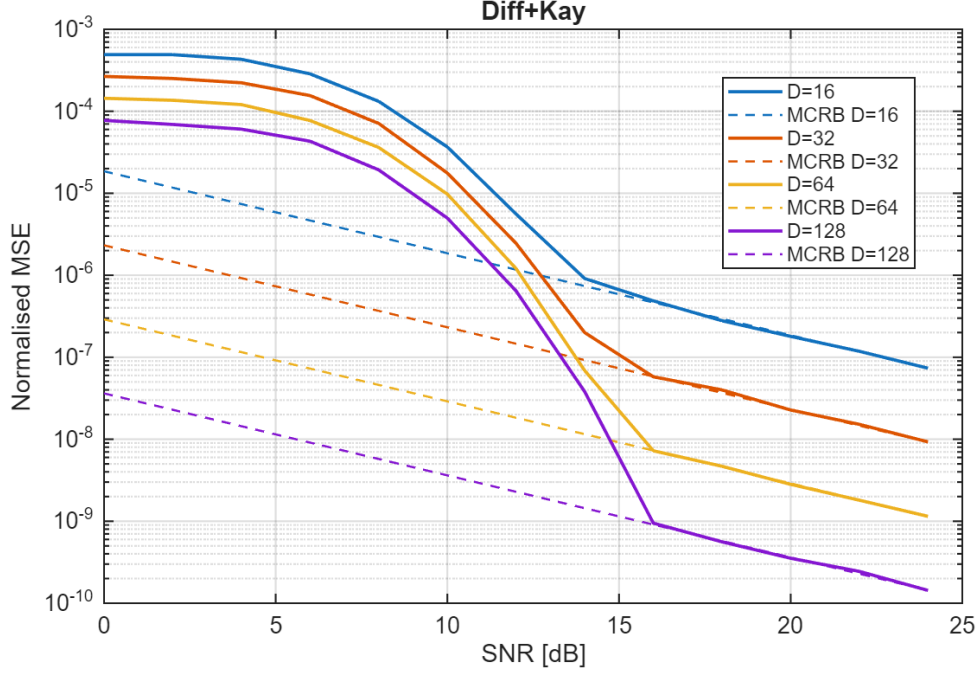


Figure 5: Normalised MSE of the DK estimator with blind operation for different observation lengths.

1.1.4 Computational Complexity

Here the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. Real division will also be assumed to cost the same as multiplication as it is only performed once (this would be a suitable approximation if look-up tables are used to find reciprocals at some cost to accuracy). All analysis is for data-aided operation, avoiding the fourth power operation in ??, and applies to a single polarization. If averaging is done over both polarizations, then all operations are doubled.

R&B The R&B algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re \left[\frac{a + bj}{c + dj} \right] = \frac{ac + bd}{c^2 + d^2}. \quad (9)$$

The total complexity is summarised in ?? with the complexity for blind operation marked with (*).

Operation	RM	RA
Forming $z[i], \dots, z[i + L]$	$4L, 12L^*$	$3L, 9L^*$
FFT	$2L \log_2(\frac{N_{FFT}}{L}) + 2N_{FFT} \log_2(L)$	$3L \log_2(\frac{N_{FFT}}{L}) + 3N_{FFT} \log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 1: Computational complexity of estimating CFO with the R&B algorithm.

DK The DK estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \dots, \arg(z[i + L])$	$2L$	L
Differential Phase	1	$L - 1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \dots, \arg(z[i + L])$	$2L$	L
Kay	L	$L - 1$

Table 2: Computational complexity of estimating CFO with the DK estimator.

1.1.5 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s . For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} when $0 \leq |f_d/R_s| \leq 1$ is the estimation range. In reality a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

1.2 Phase Recovery

1.2.1 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small angle approximations. The CPON spec sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7 the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g., 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data were used instead.

1.2.2 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of ???. It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg \left(\sum_{i=1}^N w_{ML}^*[i] z[i] \right) - \frac{\pi}{4}. \quad (10)$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied on all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON spec supplies 1 pilot symbol in every 32-symbol block that can be used to correct this [?]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{PU}$ and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{PU} = \hat{\theta}_{PU} - n\frac{\pi}{2}$.

1.2.3 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using ??? and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm 45^\circ$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

1.2.4 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in ???.

Operation	RM	RA
Forming $z[i], \dots, z[i + N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N + 1$	$3N + 2(N - 1) + 1$

Table 3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in ???.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 4: Computational complexity of Pilots-Only correction.

2 Methodology

All tests were run on fixed-point implementations of the above frequency and phase recovery algorithms with a CFO of 3 GHz (10% of the symbol rate) and a laser linewidth of 1 MHz to reflect typical operating conditions. The FFT size used in the R&B algorithm was fixed at 512. All tests were repeated 100 times with noise and data being randomly generated on each run.

2.1 Blind vs Data-Aided Operation

The threshold SNR was computed for a range of blind observation lengths and compared against data-aided performance for each combination of frequency recovery and phase recovery algorithm. All fixed-point implementations used 16 bits of precision.

2.2 Fixed-Point Precision

The threshold SNR was computed for the DK estimator (data-aided) and the R&B algorithm (data-aided and blind with D=512) followed by both Pilots-Only and Viterbi & Viterbi phase recovery. This was performed in two sweeps: first, the fixed-point precision of the phase recovery was varied while the frequency recovery precision was fixed at 16 bits; second, the frequency recovery precision was varied with the phase recovery precision fixed at 16 bits.

3 Results

3.1 Blind vs Data-Aided Operation

The decrease in threshold SNR for longer observation lengths as shown in ?? suggests that the R&B algorithm will eventually exhibit better performance in blind operation where the observation length is not limited. This is shown in ?? where a blind observation length of 40 – 50 achieves data-aided performance, and continues to improve up to a length of 256, where it plateaus. Unlike the R&B algorithm, the SNR threshold does not decrease for the DK estimator when the blind observation length increases. This is evident from ?? where the threshold SNR for blind operation remains well above that for data-aided operation.

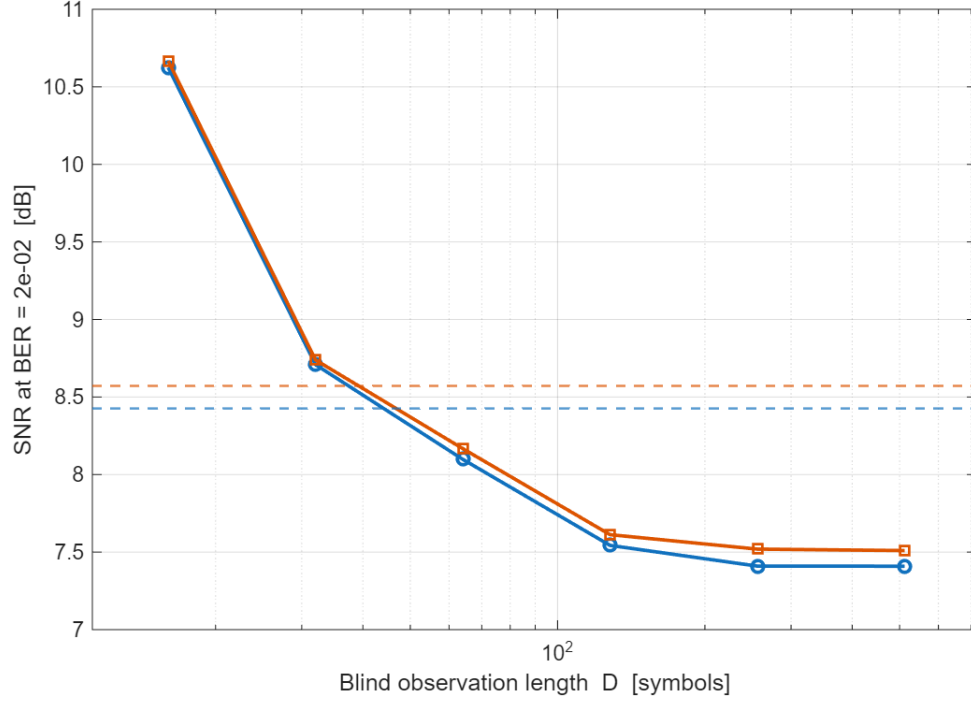


Figure 6: Threshold SNR vs blind observation length for the R&B algorithm. The dashed line is the threshold for data-aided operation. Orange: Pilots-Only. Blue: Viterbi & Viterbi.

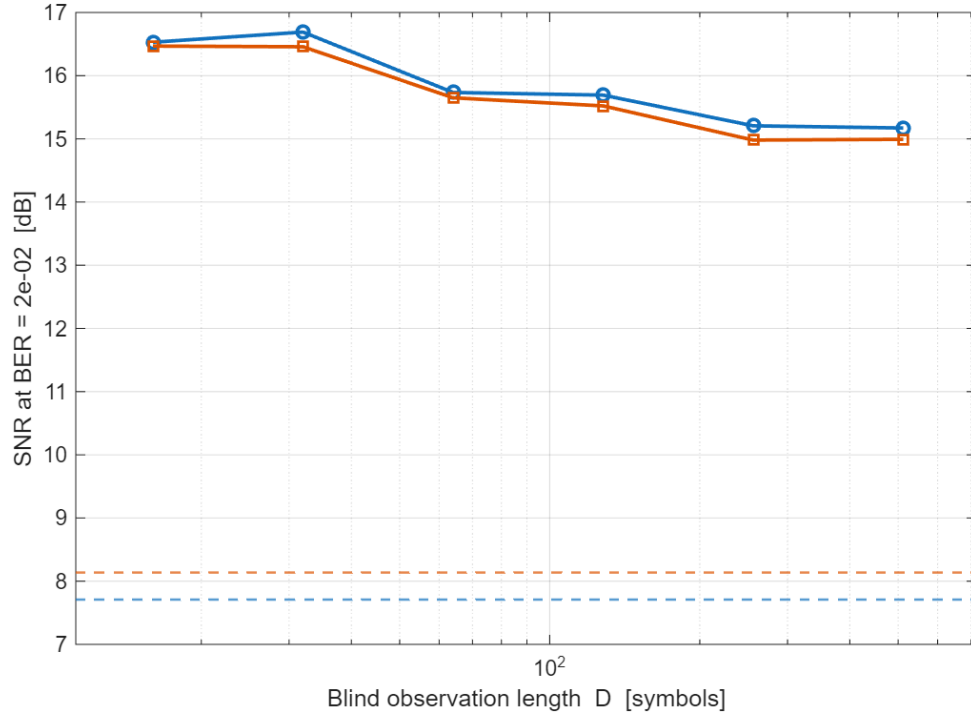


Figure 7: Threshold SNR vs blind observation length for the DK estimator. The dashed line is the threshold for data-aided operation. Orange: Pilots-Only. Blue: Viterbi & Viterbi.